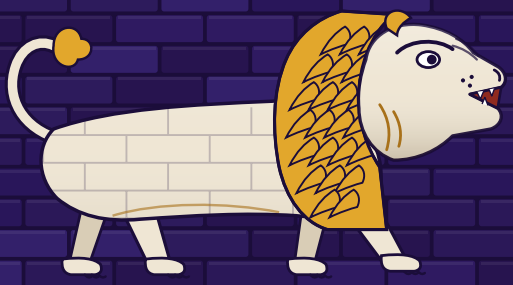
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

C1

CALCULUS I

A handout of ideas, facts and problems



12 sections · Edition of November 2026 · 38 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Calculus I

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

1 Limits, continuity and the derivative definition

Aims. Distinguish a limit from a value at the point; Find a polynomial derivative from the difference quotient.
Before this section: Expanding a square; factoring and cancelling a nonzero common factor.

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

Idea 1. A secant line measures change across separated inputs. As those inputs approach one another, its slope can approach a single instantaneous slope: the derivative. The increment must remain nonzero while it is a denominator, even though the limiting process eventually removes it.

Fact 1.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

where h is the small nonzero input increment before the limit (same unit as x); $f'(x)$ is the instantaneous rate of change (output unit per input unit).

Why it holds. Factoring and cancelling the increment can remove the apparent zero denominator. The nearby simplified slopes determine the limit, rather than an undefined quotient at zero increment.

Method 1.

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.
3. Factor out h and cancel only while h is nonzero.
4. Take the limit h approaching zero in the simplified expression.

Recognising it. If the question says use the definition, construct the difference quotient instead of citing the power rule.

Pr 1. For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$.

Solution. For $f(x) = x^2$, the incremented value is $(x+h)^2 = x^2 + 2xh + h^2$. Subtracting f(x) leaves $2xh + h^2$. Dividing by nonzero h gives $2x + h$. As h approaches zero, the secant slope approaches 2x, so $f'(x) = 2x$.

Common error. Zero over zero is an indeterminate form, not a final answer of zero.

In the examination. Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

Pr 2. Why should h not be set to zero before simplifying the difference quotient?

Pr 3. Use the definition to find the derivative of f(x)=3x squared.

Pr 4. Why can the increment be cancelled before taking the derivative limit?

(a) It is nonzero while approaching zero

(b) It equals zero at every stage

(c) Any denominator may always be cancelled

2 Basic derivatives and tangent lines

Aims. Differentiate sums and powers correctly; Build a tangent line from the original point and derivative slope.
Before this section: Power rule; substituting an x value; point-slope form of a line.

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.

Idea 2. A derivative supplies a slope, not a whole line. Many parallel lines share that slope, so a tangent must also be anchored at its point on the original curve. Calculate the curve height from the original function and the slope from its derivative.

Fact 2.

$$\frac{d}{dx}x^n = nx^{n-1}$$
$$y - f(a) = f'(a)(x - a)$$

where a is the input coordinate of tangent point (input unit); $f(a)$ is the output coordinate of tangent point (output unit); $f'(a)$ is the tangent slope (output unit per input unit).

Why it holds. The original function gives the tangent point. The derivative gives the local slope, and a point plus a finite slope determines the line.

Method 2.

1. Differentiate the original function term by term.
2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

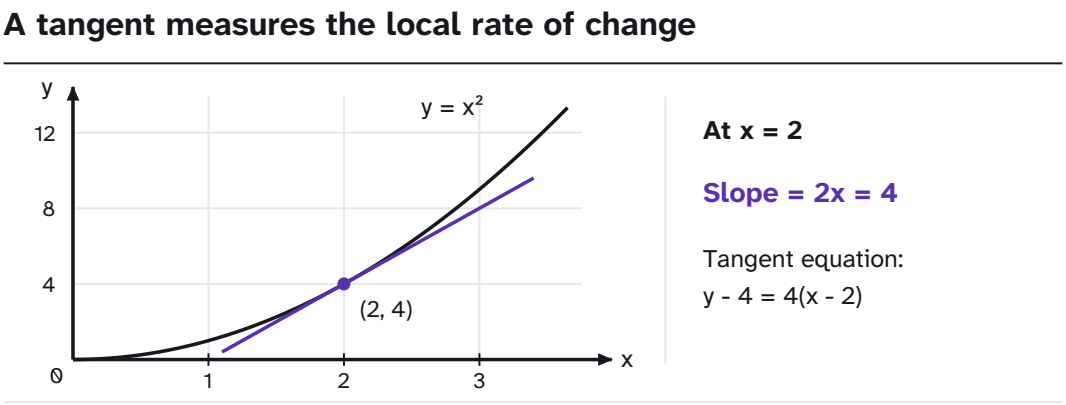


Figure 1. Figure 04. The tangent to $y = x^2$ at $(2, 4)$ has slope 4 and equation $y - 4 = 4(x - 2)$. Its plotted line touches the curve at the marked point. Read slope from coordinates, not the visual angle.

Recognising it. A tangent-equation request needs both a point and a slope, even when differentiation itself is simple.

Pr 5. For $f(x) = x^2 + 1$ at $x = 2$, the point is $(2,5)$ and slope is 4.

Solution. For $f(x) = x^2 + 1$, $f'(x) = 2x$ because the constant contributes zero. At $x = 2$ the slope is 4, while the original function gives $y = 5$. The tangent must therefore pass through $(2,5)$ with slope 4. Its equation is $y - 5 = 4(x-2)$, which expands to $y = 4x - 3$.

Common error. Do not use $f'(a)$ as the y-coordinate.

In the examination. Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

Pr 6. Does substituting $x = 2$ into f' give the y coordinate of the tangent point?

Pr 7. Find the tangent to $f(x)=x$ cubed at $x = 2$.

Pr 8. Two points have the same finite derivative value. Their tangent lines must be:

(a) Exactly the same line in every case

(b) Parallel or coincident

(c) Perpendicular

3 Product and quotient rules

Aims. Differentiate a product using both changing factors; Preserve term order and the squared denominator in the quotient rule.
Before this section: Basic polynomial and sine derivatives; multiplication and fractions.

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

IDEA 3. When two factors change, a product changes through both of them. A quotient also has two changing parts, but its denominator contribution enters with a subtraction. Writing the full rule before simplifying preserves the separate contributions and protects the sign.

FACT 3.

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

where u, v is the the two factor functions (problem dependent); u', v' is the their derivatives with respect to x (corresponding output per input unit).

WHY IT HOLDS. The product rule includes one contribution from each changing factor. Reversing the quotient-rule subtraction reverses the final derivative sign.

METHOD 3.

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

RECOGNISING IT. Inspect the outer operation: multiplication or division of functions signals the corresponding rule.

Pr 9. For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$.

Solution. For y = x^2 sin x, choose u = x^2 and v = sin x. Their derivatives are u′ = 2x and v′ = cos x. The product rule keeps each factor once unchanged: u′v + uv′. Thus y′ = 2x sin x + x^2 cos x; multiplying the two derivatives alone would miss both contributions.

Common error. The quotient denominator is the square of the original denominator.

In the examination. Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

Pr 10. For $y = (x+1)/(x-1)$, what is the derivative for $x \neq 1$?

Pr 11. Differentiate $y=(x+1)/(x-1)$, for x not equal to 1.

Pr 12. Why is separately differentiating top and bottom generally wrong for the derivative of u/v?

(a) The denominator must always be constant

(b) Fractions cannot be differentiated

(c) It omits the complete effect of denominator change

4 Chain rule and trigonometric functions

Aims. Identify inner and outer operations in a composite function; Include the inner derivative in trigonometric and power derivatives.
Before this section: Power rule; sine and cosine derivatives; radians.

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

Idea 4. A composite function is a machine inside a machine. The outer function responds to its inner input, while that inner input changes with the original variable. The chain rule multiplies these rates to describe the complete change.

Fact 4.

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$
$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$

where $u = g(x)$ is the inner expression (problem dependent); $g'(x)$ is the inner derivative (inner-expression unit per input unit).

Why it holds. An outer derivative alone measures change per unit of inner expression. The inner derivative converts that rate into change per unit of the original variable.

Method 4.

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.
4. Check the sign for cosine derivatives and use radians for trigonometric rules.

Recognising it. An operation surrounding a nontrivial expression, such as cosine surrounding a square, is a chain-rule cue.

Pr 13. For $y = (3x + 1)^4$, derivative $= 4(3x + 1)^3 \cdot 3 = 12(3x + 1)^3$.

Solution. In y = $(3x+1)^4$, the inner expression is 3x+1 and the outer operation is raising to the fourth power. The outer derivative gives $4(3x+1)^3$. The inner derivative is 3, so it must multiply the outer result. Combining the constants gives $y' = 12(3x+1)^3$.

Common error. Forgetting the inner derivative is the most common lost mark.

In the examination. Before finishing, circle the inner expression and verify that its derivative appears as a factor.

Pr 14. What is the derivative of $\cos(x^2)$?

Pr 15. Differentiate $y=\cos(x^2)$.

Pr 16. In differentiating $\sin(7x)$, what does the extra factor 7 represent?

(a) The rate of change of the inner expression

(b) A second sine factor

(c) A degrees-to-radians conversion

Fact 5. Four more trigonometric derivatives. The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

$$\frac{d}{dx}\tan u = \sec^2 u\, u'$$

$$\frac{d}{dx}\cot u = -\csc^2 u\, u'$$

$$\frac{d}{dx}\sec u = \sec u \tan u\, u'$$

$$\frac{d}{dx}\csc u = -\csc u \cot u\, u'$$

1. Use radians. Here u is a function of x, and u' means its derivative.
2. Apply the outside rule, then multiply by the inside derivative.
3. Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.

Pr 17. Differentiate $y = \cot(5x)$.

5 Implicit differentiation

Aims. Differentiate y terms while treating y as a function of x; Collect and isolate all terms containing y'.
Before this section: Chain and product rules; algebraic rearrangement.

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces y'. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

Idea 5. An implicit equation links two changing coordinates without isolating one globally. As one coordinate changes, the other adjusts to stay on the curve. Differentiation records this linked movement, so differentiating a function of the dependent coordinate requires its own rate of change.

Fact 6.

$$\frac{d}{dx}(y^2) = 2yy'$$
$$\frac{d}{dx}(xy) = y + xy'$$

where y' is the dy/dx, the local slope of the curve (y unit per x unit).

Why it holds. A y-expression is a composite function of x, so chain-rule factors appear. Collecting derivative terms reveals a shared factor that can be isolated to find the slope.

Method 5.

1. Differentiate both sides with respect to x.
2. Attach y' when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y' to one side.
4. Factor out y', divide by its coefficient and state where that division is valid.

Recognising it. Use implicit differentiation when x and y are mixed and explicit rearrangement would be awkward or lose branches.

Pr 18. From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$.

Solution. The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value. Differentiating x^2 gives 2x; differentiating y^2 gives 2y y' by the chain rule. The constant 25 differentiates to zero, so $2x + 2y\,y' = 0$. Rearranging gives $y' = -x/y$ wherever $y \neq 0$; division by y requires this restriction.

Common error. Do not treat y as a constant or omit the product rule on xy.

In the examination. For an xy term write $y + xy'$, not simply y or xy'; both factors depend on x.

Pr 19. Why does the derivative of y^2 contain y'?

Pr 20. Find dy/dx for xy+y squared=6.

Pr 21. When y depends on x, the derivative of xy is:

(a) xy'

(b) $y + xy'$

(c) y

6 Logarithmic, exponential and inverse-trig derivatives

Aims. Differentiate exponential and logarithmic composites; Use logarithmic differentiation for a positive variable base and exponent.
Before this section: Chain rule; natural logarithms; product rule; positive-domain restrictions.

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

Idea 6. A variable power changes through both its base and its exponent. A logarithm moves the exponent into a product, where familiar differentiation rules track those two changes. The final step restores the original function after differentiating the logarithmic equation.

Fact 7.

$$(e^u)' = e^u u'$$
$$(\ln|u|)' = u'/u$$
$$(\arctan x)' = 1/(1+x^2)$$

where e is the base of the natural exponential (dimensionless); $\ln u$ is the natural logarithm of a positive dimensionless argument (dimensionless).

Why it holds. Logarithms turn a positive power expression into a product involving its exponent. The derivative of the logarithm includes the original derivative divided by the original function, so that function must be restored.

Method 6.

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.
4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

Recognising it. A variable in both base and exponent suggests logarithmic differentiation, after checking a suitable positive domain.

Pr 22. For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$.

Solution. For ln(x^2+1), the inner derivative is 2x and the inner expression is always positive. The logarithmic chain rule therefore gives 2x/(x^2+1). For the separate function e^(3x), the exponential is retained unchanged. Multiplication by the inner derivative 3 gives 3e^(3x).

Common error. The derivative of ln u is u prime/u, not 1/ln u.

In the examination. The power rule nx^(n−1) assumes a constant exponent; x^x needs another method.

Pr 23. For x > 0, what is d(x^x)/dx?

Pr 24. Differentiate y=x to the power x for x greater than zero.

Pr 25. Why is the constant-exponent power rule insufficient for x^x?

(a) The expression is constant

(b) Logarithms are invalid for positive x

(c) The exponent changes as well as the base

Fact 8. Other exponential, logarithm and inverse-trig rules. The supplied final includes 3^x and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

$$\frac{d}{dx}a^u = a^u \ln(a)\, u'$$

$$\frac{d}{dx}\log_a u = \frac{u'}{u \ln a}$$

$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$$

1. The base a is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
2. For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
3. Find the inside derivative u' first. Inverse sine means arcsin; it is different from 1/ sin.

Pr 26. Differentiate 3^x and $\log_3 x$.

7 L Hopital and indeterminate forms

Aims. Check for a permitted indeterminate quotient before using L Hopital; Evaluate a simple limit by differentiating numerator and denominator.
Before this section: Limits; exponential and trigonometric derivatives; chain rule.

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

Idea 7. An indeterminate quotient describes competition between quantities approaching zero or growing without bound. When its conditions hold, L Hopital compares their rates of change. It is a limit procedure, so it does not replace the quotient rule for the derivative of a fraction.

Fact 9.

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

where $0/0$ is the indeterminate limiting form, not an arithmetic value (not a measured quantity); ∞/∞ is the indeterminate quotient form (not a measured quantity).
Conditions. The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

Why it holds. The initial limiting form must be checked before applying the rule. Differentiating a numerator containing an inner expression still requires the chain rule.

Method 7.

1. Substitute the limiting input to identify the quotient form.
2. Confirm $0/0$ or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

Recognising it. Check the quotient form first; an ordinary fraction alone is not a reason to use L Hopital.

Pr 27. As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$.

Solution. As x approaches zero, e^x - 1 approaches zero and x approaches zero. The quotient therefore has the required $0/0$ form. Differentiate numerator and denominator separately to obtain e^x/1. Its limit is e^0 = 1, so the original limit is 1.

Common error. Do not use the rule on a quotient whose substitution gives an ordinary finite value.

In the examination. Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

Pr 28. What is the limit of $\sin(3x)/x$ as x approaches zero?

Pr 29. Evaluate the limit of $\sin(3x)/x$ as x approaches zero.

Pr 30. A numerator approaches 2 and its denominator approaches 4. The appropriate first method is:

(a) Use the ordinary quotient limit

(b) Apply L Hopital because it is a fraction

(c) Declare the form indeterminate

8 Increasing, decreasing and concavity

Aims. Use a first-derivative sign chart to find monotonic intervals; Distinguish a stationary point from a proven extremum or inflection.
Before this section: Factoring quadratics; sign tests; first and second derivatives.

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

Idea 8. The derivative records the local slope of the graph. A negative derivative makes the graph fall as input increases; a positive derivative makes it rise. A change from falling to rising identifies a minimum, whereas a zero derivative without a sign change may not be an extremum.

Fact 10.

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

where f' is the first derivative, controlling increase or decrease (output per input unit); f'' is the second derivative, controlling concavity (output per input unit squared).

Why it holds. A zero derivative gives a candidate rather than a complete extremum proof. Signs on both sides tell whether increasing or decreasing behaviour actually changes.

Method 8.

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.
4. Use f′′ signs separately for concavity and require a concavity change for an inflection point.

Recognising it. Use a derivative sign chart for monotonic intervals and local extrema; use a separate concavity test for inflections.

Pr 31. For $f = x^3 - 3x$, derivative is $3(x-1)(x+1)$.

Solution. For f(x) = x³ - 3x, f′ = 3x² - 3 = 3(x−1)(x+1). The critical numbers are −1 and 1. Testing x = −2, 0 and 2 gives derivative signs positive, negative and positive. The function increases on (-∞,−1) and (1,∞), and decreases on (−1,1); the sign changes identify a local maximum at −1 and minimum at 1.

Common error. A derivative equal to zero does not automatically mean a maximum.

In the examination. Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

Pr 32. Is $f''(c) = 0$ alone enough to prove an inflection point?

Pr 33. For $f(x) = x^2 - 4x$, find increasing and decreasing intervals and the minimum point.

Pr 34. A first derivative changes from negative to positive at an interior critical point. This gives:

(a) An inflection point guaranteed

(b) A local minimum

(c) A local maximum

9 Graphing, asymptotes and absolute extrema

Aims. Find absolute extrema by comparing all candidates on a closed interval; Distinguish a removable hole from a vertical asymptote.
Before this section: Domain restrictions; critical numbers; evaluating a function at endpoints.

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

Idea 9. An absolute extremum compares the entire allowed graph. On a closed interval, an endpoint can be highest or lowest without having a horizontal tangent. Comparing a short list of original function values turns this global problem into a finite check.

Fact 11.

$$\text{Candidates} = \text{endpoints plus interior critical numbers}$$

where $[a, b]$ is the closed interval including both endpoints (input unit); $f(c)$ is the function value at a candidate point (output unit).

Why it holds. Interior critical numbers supply possible internal extrema. Endpoints must also be considered because the allowed domain ends there.

Method 9.

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.
4. Compare the values and report each extremum value together with where it occurs.

Recognising it. Absolute maximum or minimum on a closed interval calls for endpoint-plus-critical-value comparison.

Pr 35. For x^2 on $[-1, 2]$, candidates are -1,0,2.

Solution. For f(x) = x^2 on [−1,2], the derivative 2x is zero at x = 0. The candidate inputs are the two endpoints −1 and 2 and the interior critical number 0. The original function values are f(−1)=1, f(0)=0 and f(2)=4. The smallest is 0 at x=0; the largest is 4 at x=2.

Common error. Do not omit endpoints or compare derivative values instead of function values.

In the examination. Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

Pr 36. Can an endpoint be the absolute maximum even when its derivative is not zero?

Pr 37. Find absolute maximum and minimum of $f(x) = x^2-2x$ on [0,3].

Pr 38. A continuous function is strictly increasing on a closed interval. Its absolute maximum occurs:

(a) Only where its derivative is zero

(b) Nowhere because it keeps increasing

(c) At the right endpoint

10 Optimization and mean-value theorems

Aims. Translate a simple optimization constraint into one variable; Check mean-value-theorem assumptions and solve for the interior input.

Before this section: Area and perimeter; derivatives; continuity and differentiability.

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

Idea 10. The mean-value theorem links average change across an interval with instantaneous change somewhere inside. A secant joins endpoint heights, and at least one interior tangent must be parallel under the required conditions. The theorem matches slopes rather than automatically selecting a midpoint.

Fact 12.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

where x is the one rectangle side length (length unit); A is the rectangle area (square length unit); c is the interior input guaranteed by the mean-value theorem (input unit).

Why it holds. Continuity and interior differentiability exclude breaks and corners that could defeat the guarantee. The matching input lies strictly inside the interval but need not be its midpoint.

Method 10.

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.
4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

Recognising it. Use the theorem when a function and interval are supplied and an interior slope-matching input is requested.

Pr 39. A rectangle with perimeter 20 has area $x(10 - x)$.

Solution. Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x. Area becomes A(x) = x(10-x) with $0 < x < 10$ for a nondegenerate rectangle. Its derivative is $10 - 2x$, which vanishes at x = 5. The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

Common error. A candidate outside the physical domain is not a valid optimization answer.

In the examination. A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

Pr 40. For $f(x)=x^2$ on [1,3], what c satisfies the mean-value theorem?

Pr 41. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

Pr 42. Must the input guaranteed by the mean-value theorem be the midpoint?

(a) No; it solves the slope-matching equation

(b) Yes; every function uses the midpoint

(c) It must be an endpoint

11 Antiderivatives and substitution

Aims. Choose a substitution whose differential matches the integrand; Return the antiderivative to x and include the integration constant.

Before this section: Chain rule; power integration; constant factors.

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

IDEA 11. Substitution recognizes the chain rule running backward. A temporary variable replaces an inner expression, while its differential accounts for a matching factor elsewhere. A leftover constant is harmless if tracked carefully rather than dropped.

FACT 13.

$$\int x^n\,dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

where u is the chosen inner expression (problem dependent); du is the differential of the inner expression (same unit as u); C is the arbitrary additive integration constant (antiderivative unit).

WHY IT HOLDS. A useful substitution replaces the nested expression and its differential factor together. Differentiating the final result checks both the power and its multiplier.

METHOD 11.

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.
4. Substitute back to x, add C and differentiate once to check.

RECOGNISING IT. Look for an inner expression accompanied by its derivative or a constant multiple of that derivative.

Pr 43. For the integral of $2x(x^2+1)^3$, set $u = x^2+1$ and $du = 2x\,dx$.

Solution. For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$. Then $du = 2x\,dx$, so the complete integral becomes the integral of $u^3\,du$. The antiderivative is $u^4/4 + C$. Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

Common error. The power rule exception is $n=-1$; do not divide by zero.

In the examination. Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

Pr 44. What is the integral of $6x(x^2+2)^2$ dx?

Pr 45. Evaluate the integral of $6x(x^2+2)$ squared with respect to x.

Pr 46. Which substitution most directly simplifies $6x(x^2+2)^2$?

(a) Rename the whole integrand without computing a differential

(b) $u = x^2 + 2$

(c) $u = 6x$

Fact 14. Integrate sine, cosine and an exponential. These short rules reverse the chain rule. A sine integral appears in the supplied final.

$$\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + C$$

$$\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + C$$

$$\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + C$$

1. Here a and b are constants and $a \neq 0$. Trigonometric angles use radians.
2. Divide by the coefficient a; differentiation would multiply by it.
3. Add C for an indefinite integral. Differentiate your answer to check it.

Pr 47. Find $\int \sin(3x)\,dx$.

12 Definite integrals, signed area and the fundamental theorem

Aims. Distinguish signed integral from total geometrical area; Apply the fundamental theorem, including a variable-upper-limit chain factor.
Before this section: Antiderivatives; positive and negative regions; substituting bounds.

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

Idea 12. A definite integral combines signed contributions over fixed bounds. An antiderivative provides a shortcut through its change between endpoints. If the integrand is nonnegative throughout, the signed accumulation is also geometric area; otherwise positive and negative regions can cancel.

Fact 15.

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

where a, b is the lower and upper integration bounds (input unit); F is the an antiderivative of f (f unit times input unit); t is the dummy integration variable (input unit).

Why it holds. The fundamental theorem requires upper antiderivative value minus lower value. Substitution changes the endpoint labels unless the antiderivative is returned to the original variable before evaluation.

Method 12.

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

Recognising it. With fixed limits, keep endpoint values consistent with the integration variable and evaluate their difference.

Pr 48. The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1.

Solution. For y = x from -1 to 1, the region below the axis contributes negative signed area. An antiderivative is $x^2/2$; evaluation at 1 and -1 gives $1/2 - 1/2 = 0$. Geometrically each half is a triangle with area 1/2. Total geometrical area is $1/2 + 1/2 = 1$, while the signed integral remains zero.

Common error. Never keep x-limits on an integral that has been fully changed to u.

In the examination. For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

Pr 49. Does a zero definite integral prove the function is zero throughout the interval?

Pr 50. Evaluate the integral from 0 to 1 of $2x(x^2+1)$ dx.

Pr 51. After changing variable in a definite integral, what should happen to the bounds?

(a) Always keep the old bound numbers with the new variable

(b) Remove them permanently

(c) Transform them, or return to the original variable before evaluation

FACT 16. Read a Riemann sum as rectangles. The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

$$\Delta x = \frac{b-a}{n}$$

$$\text{rectangle sum} = \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

1. Split the interval from a to b into n equal widths.
2. Choose a sample point x_i^* in each small interval. Its function value is that rectangle's signed height.
3. Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
4. Below-axis heights contribute negative signed area.

Pr 52. Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Write $f(x+h) = 3(x+h)^2$, then subtract the complete original function. (2) Expand and factor h from the numerator of the difference quotient. (3) Cancel nonzero h, then let it approach zero. Keep the coefficient 3 throughout.

Pr 7 (p. 4) (1) Evaluate x^3 at $x = 2$ to find the point on the original curve. (2) Differentiate x^3 and evaluate the derivative at the same input. (3) Use the point and slope in $y - y_0 = m(x - x_0)$, then check that the point satisfies your line.

Pr 11 (p. 6) (1) Choose $u = x + 1$ and $v = x - 1$, then find both derivatives. (2) Form $(u'v - uv')/v^2$ with brackets around both numerator terms. (3) Simplify $(x-1) - (x+1)$ carefully and preserve $x \neq 1$.

Pr 15 (p. 7) (1) Identify cosine as the outer function and x^2 as the inner expression. (2) Differentiate cosine while leaving x^2 inside, and retain the negative sign. (3) Multiply by the derivative of x^2. Check that both rates appear.

Pr 17 (p. 7) (1) The inside is $u = 5x$, so $u' = 5$. (2) The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

Pr 20 (p. 9) (1) Differentiate both sides of $xy + y^2 = 6$, using the product rule for xy. (2) Include y' when differentiating y^2, and collect all derivative terms. (3) Factor out y', move the remaining term and divide by its coefficient where nonzero.

Pr 24 (p. 10) (1) For $x > 0$, set $y = x^x$ and take natural logarithms. (2) Rewrite the right side as $x \ln x$; use the product rule while the left derivative is y'/y. (3) Multiply by y, then replace it with the original x^x.

Pr 26 (p. 10) (1) In both cases the inside expression is $u = x$, so $u' = 1$. (2) For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

Pr 29 (p. 12) (1) Check the numerator and denominator as $x \to 0$ and identify the form. (2) Differentiate $\sin(3x)$ with its inner derivative and differentiate the denominator separately. (3) Evaluate the resulting derivative quotient at the limiting input; do not use a quotient-rule numerator.

Pr 33 (p. 13) (1) Differentiate $x^2 - 4x$ and find where its derivative is zero. (2) Test derivative signs using one input on each side of that critical number. (3) Report the resulting intervals and evaluate the original function at the critical input to give the minimum point.

Pr 37 (p. 14) (1) Find the derivative of $x^2 - 2x$ and its critical input inside $[0, 3]$. (2) List that input and both endpoints as candidates. (3) Evaluate the original function at each candidate; report both the largest and smallest values and where each occurs.

Pr 41 (p. 15) (1) Check that x^2 is continuous on $[1, 3]$ and differentiable inside. (2) Compute the endpoint secant slope from function values and interval length. (3) Set $f'(c)$ equal to that slope, solve and verify $1 < c < 3$.

Pr 45 (p. 17) (1) Choose $u = x^2 + 2$, calculate du and compare it with $6x\,dx$. (2) Rewrite the full integral in u, retaining any constant multiplier. (3) Integrate the power, return to x, include the arbitrary constant and differentiate to check.

Pr 47 (p. 17) (1) The sine antiderivative is negative cosine. (2) The inside derivative is 3, so divide by 3. (3) Check: $\dfrac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

Pr 50 (p. 18) (1) Notice $x^2 + 1$ and its derivative factor $2x\,dx$. (2) Substitute with transformed bounds, or find an antiderivative and return it to x. (3) Evaluate upper minus lower between 0 and 1. Since the integrand is nonnegative there, a negative result would indicate an error.

Pr 52 (p. 19) (1) Each width is $\Delta x = (1 - 0)/2 = 1/2$. (2) The right endpoints are $1/2$ and 1, giving heights $1/2$ and 1. (3) The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Answers

Pr 2 (p. 3) It would produce division by zero; the limit concerns nonzero h approaching zero.

Pr 3 (p. 3) [3(x+h) squared −3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Pr 4 (p. 3) (a) Correct. Simplification is valid at nonzero increments, and those nearby values determine the limit.

Pr 6 (p. 4) No. It gives the slope; use f(2) for the y coordinate.

Pr 7 (p. 4) f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y−8=12(x−2), equivalently y=12x-16. One mark per step.

Pr 8 (p. 4) (b) Correct. Equal slopes give parallel lines, which can coincide in a special case.

Pr 10 (p. 6) $-2/(x-1)^2$: the numerator is (x−1) − (x+1) = −2.

Pr 11 (p. 6) Derivative = [(x−1)-(x+1)]/(x−1) squared = −2/(x−1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Pr 12 (p. 6) (c) Correct. The quotient rule accounts for both changes and includes the squared denominator.

Pr 14 (p. 7) $-2x \sin(x^2)$, combining the cosine derivative with the inner derivative 2x.

Pr 15 (p. 7) Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain $-2x \sin(x^2)$. Credit both layers, sign and product.

Pr 16 (p. 7) (a) Correct. The inner expression changes seven units per unit increase in x.

Pr 17 (p. 7) $y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

Pr 19 (p. 9) Because y depends on x; the chain rule multiplies 2y by dy/dx.

Pr 20 (p. 9) Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Pr 21 (p. 9) (b) Correct. Both factors change, so the product rule gives both terms.

Pr 23 (p. 10) x^x(ln x + 1). Take ln y = x ln x, then $y'/y = \ln x + 1$.

Pr 24 (p. 10) Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x × (ln x+1). Four marks for these stages.

Pr 25 (p. 10) (c) Correct. Treating the exponent as fixed misses part of the change.

Pr 26 (p. 10) $(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

Pr 28 (p. 12) 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

Pr 29 (p. 12) The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Pr 30 (p. 12) (a) Correct. A nonzero finite denominator limit gives a directly determined quotient.

Pr 32 (p. 13) No. Concavity must change across c.

Pr 33 (p. 13) Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,−4). One mark each for derivative, intervals and point.

Pr 34 (p. 13) (b) Correct. The graph falls toward the point and rises away.

Pr 36 (p. 14) Yes. Absolute extrema on a closed interval must include endpoint checks.

Pr 37 (p. 14) Critical point x=1. Values: f(0)=0, f(1)=−1, f(3)=3. Minimum −1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Pr 38 (p. 14) (c) Correct. The last allowed input has the largest output.

Pr 40 (p. 15) c=2: the average slope is (9−1)/(3−1)=4 and f′(c)=2c=4.

Pr 41 (p. 15) The polynomial is continuous and differentiable. Average slope = (9−1)/(3−1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Pr 42 (p. 15) (a) Correct. A midpoint works for some functions but is not the general theorem conclusion.

Pr 44 (p. 17) $(x^2+2)^3 + C$. With $u=x^2+2$, the integral becomes 3 times the integral of u^2 du.

Pr 45 (p. 17) Set $u=x^2+2$, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Pr 46 (p. 17) (b) Correct. Its differential matches the remaining x factor up to a constant.

Pr 47 (p. 17) $-\frac{1}{3}\cos(3x) + C$.

Pr 49 (p. 18) No. Positive and negative contributions can cancel.

Pr 50 (p. 18) Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4−1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Pr 51 (p. 18) (c) Correct. The endpoints must still describe the same original inputs.

Pr 52 (p. 19) Estimate: 3/4. Exact integral: 1/2. The right-end estimate is too large because the function increases.

Calculus I, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. The derivative of x^2 at $x = 3$ is:

   (a) 3
   (b) 6
   (c) 9
   (d) 0

2. The derivative of a constant 7 is:

   (a) 7
   (b) 1
   (c) x
   (d) 0

3. The derivative of xe^x is:

   (a) $e^x(1 + x)$
   (b) 0
   (c) e^x
   (d) xe^x

4. The derivative of $\sin(5x)$ is:

   (a) $-5\sin(5x)$
   (b) $\sin(5)$
   (c) $\cos(5x)$
   (d) $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at $(3,4)$ is:

   (a) 0
   (b) $\frac{3}{4}$
   (c) $-\frac{3}{4}$
   (d) $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

   (a) $\arctan(1)$
   (b) $\frac{1}{1+x^2}$
   (c) $\frac{1}{\cos^2 x}$
   (d) $-\frac{1}{1+x^2}$

Part II. Show your reasoning.

1. Use the definition to find the derivative of f(x)=3x squared.
2. Find the tangent to f(x)=x cubed at $x = 2$.
3. Differentiate y=(x+1)/(x−1), for x not equal to 1.
4. Differentiate y=cos(x^2).
5. Find dy/dx for xy+y squared=6.
6. Differentiate y=x to the power x for x greater than zero.

Calculus I, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Before applying L Hopital directly, the quotient should have form:

   (a) $\frac{0}{0}$ or $\frac{\infty}{\infty}$
   (b) $\frac{1}{1}$ only
   (c) $\frac{0}{5}$ only
   (d) Any fraction

2. If f' is negative on an interval, f is:

   (a) Decreasing
   (b) Constant
   (c) Always concave down
   (d) Increasing

3. To find absolute extrema on a closed interval, check:

   (a) Only where f' is zero
   (b) Only endpoints
   (c) Endpoints and interior critical numbers
   (d) Only where f is zero

4. Rolle theorem additionally requires:

   (a) An infinite interval
   (b) Equal endpoint function values
   (c) A discontinuity
   (d) A nonzero derivative everywhere

5. An indefinite integral must generally include:

   (a) Only a numerical answer
   (b) A fixed upper limit
   (c) No variable
   (d) An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

   (a) 8
   (b) -4
   (c) 0
   (d) 4

Part II. Show your reasoning.

1. Evaluate the limit of $\sin(3x)/x$ as x approaches zero.
2. For $f(x) = x^2 - 4x$, find increasing and decreasing intervals and the minimum point.
3. Find absolute maximum and minimum of $f(x) = x^2 - 2x$ on $[0,3]$.
4. Use the mean-value theorem for $f(x) = x^2$ on $[1,3]$ to find c.
5. Evaluate the integral of $6x(x^2+2)$ squared with respect to x.
6. Evaluate the integral from 0 to 1 of $2x(x^2+1)$ dx.

Calculus I, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The derivative is 2x, so its value at 3 is 6.

Written part. [3(x+h) squared −3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit. (Section: p. 2.)

Question 2. (d) A constant function has zero rate of change.

Written part. f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y−8=12(x−2), equivalently y=12x-16. One mark per step. (Section: p. 3.)

Question 3. (a) The product rule differentiates each factor in turn.

Written part. Derivative = [(x−1)-(x+1)]/(x−1) squared = −2/(x−1) squared. Award marks for quotient rule, numerator order, denominator square and simplification. (Section: p. 5.)

Question 4. (d) The inner derivative contributes a factor of 5.

Written part. Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain −2x sin(x^2). Credit both layers, sign and product. (Section: p. 6.)

Question 5. (c) Implicit differentiation gives slope $-\frac{x}{y}$.

Written part. Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps. (Section: p. 8.)

Question 6. (b) This is the standard inverse-tangent derivative.

Written part. Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x × (ln x+1). Four marks for these stages. (Section: p. 9.)

Calculus I, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) The rule concerns specified indeterminate quotient forms.

Written part. The form is $\frac{0}{0}$. L Hopital gives limit $3\cos(3x)/1 = 3$. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer. (Section: p. 11.)

Question 2. (a) Negative slope means the function decreases as x increases.

Written part. Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,−4). One mark each for derivative, intervals and point. (Section: p. 12.)

Question 3. (c) The largest or smallest value may occur at either kind of candidate.

Written part. Critical point x=1. Values: f(0)=0, f(1)=−1, f(3)=3. Minimum −1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema. (Section: p. 13.)

Question 4. (b) Equal endpoint heights make the average slope zero.

Written part. The polynomial is continuous and differentiable. Average slope = (9−1)/(3−1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check. (Section: p. 14.)

Question 5. (d) Differentiation loses constants, so all antiderivatives differ by a constant on an interval.

Written part. Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant. (Section: p. 16.)

Question 6. (c) The odd integrand has equal and opposite signed areas.

Written part. Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4−1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction. (Section: p. 17.)

Facts at a glance

1. Limits, continuity and the derivative definition p. 2

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

2. Basic derivatives and tangent lines p. 3

$$\frac{d}{dx} x^n = n x^{n-1}$$
$$y - f(a) = f'(a)(x - a)$$

3. Product and quotient rules p. 5

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

4. Chain rule and trigonometric functions p. 6

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$
$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$

5. Implicit differentiation p. 8

$$\frac{d}{dx}(y^2) = 2yy'$$
$$\frac{d}{dx}(xy) = y + xy'$$

6. Logarithmic, exponential and inverse-trig derivatives p. 9

$$(e^u)' = e^u u'$$
$$(\ln|u|)' = u'/u$$
$$(\arctan x)' = 1/(1 + x^2)$$

7. L Hopital and indeterminate forms p. 11

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$
when the rule applies

8. Increasing, decreasing and concavity p. 12

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

9. Graphing, asymptotes and absolute extrema p. 13

Candidates = endpoints plus interior critical numbers

10. Optimization and mean-value theorems p. 14

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

11. Antiderivatives and substitution p. 16

$$\int x^n \, dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$
$$\int dx/x = \ln|x| + C$$

12. Definite integrals, signed area and the fundamental theorem p. 17

$$\int_a^b f(x) \, dx = F(b) - F(a)$$
$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Limits, continuity and the derivative definition

h	small nonzero input increment before the limit (same unit as x)
$f'(x)$	instantaneous rate of change (output unit per input unit)

2. Basic derivatives and tangent lines

a	input coordinate of tangent point (input unit)
$f(a)$	output coordinate of tangent point (output unit)
$f'(a)$	tangent slope (output unit per input unit)

3. Product and quotient rules

u, v	the two factor functions (problem dependent)
u', v'	their derivatives with respect to x (corresponding output per input unit)

4. Chain rule and trigonometric functions

$u = g(x)$	inner expression (problem dependent)
$g'(x)$	inner derivative (inner-expression unit per input unit)

5. Implicit differentiation

y'	dy/dx, the local slope of the curve (y unit per x unit)

6. Logarithmic, exponential and inverse-trig derivatives

e	base of the natural exponential (dimensionless)
$\ln u$	natural logarithm of a positive dimensionless argument (dimensionless)

7. L Hopital and indeterminate forms

$0/0$	indeterminate limiting form, not an arithmetic value (not a measured quantity)
∞/∞	indeterminate quotient form (not a measured quantity)

8. Increasing, decreasing and concavity

f'	first derivative, controlling increase or decrease (output per input unit)
f''	second derivative, controlling concavity (output per input unit squared)

9. Graphing, asymptotes and absolute extrema

$[a, b]$	closed interval including both endpoints (input unit)
$f(c)$	function value at a candidate point (output unit)

10. Optimization and mean-value theorems

x	one rectangle side length (length unit)
A	rectangle area (square length unit)
c	interior input guaranteed by the mean-value theorem (input unit)

11. Antiderivatives and substitution

u	chosen inner expression (problem dependent)
du	differential of the inner expression (same unit as u)
C	arbitrary additive integration constant (antiderivative unit)

12. Definite integrals, signed area and the fundamental theorem

a, b	lower and upper integration bounds (input unit)
F	an antiderivative of f (f unit times input unit)
t	dummy integration variable (input unit)

Sources

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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Index of notation

Latin letters first, then Greek letters, then differences (Δ). The same letter can name different quantities in different sections.

$[a, b]$ closed interval including both endpoints, 13

a input coordinate of tangent point, 3

a, b lower and upper integration bounds, 18

A rectangle area, 15

c interior input guaranteed by the mean-value theorem, 15

C arbitrary additive integration constant, 16

du differential of the inner expression, 16

e base of the natural exponential, 9

f' first derivative, controlling increase or decrease, 12

f'' second derivative, controlling concavity, 12

$f'(a)$ tangent slope, 3

$f'(x)$ instantaneous rate of change, 2

$f(a)$ output coordinate of tangent point, 3

$f(c)$ function value at a candidate point, 13

F an antiderivative of f, 18

$g'(x)$ inner derivative, 6

h small nonzero input increment before the limit, 2

∞/∞ indeterminate quotient form, 11

$\ln u$ natural logarithm of a positive dimensionless argument, 9

t dummy integration variable, 18

u chosen inner expression, 16

u', v' their derivatives with respect to x, 5

u, v the two factor functions, 5

$u = g(x)$ inner expression, 6

x one rectangle side length, 15

y' dy/dx, the local slope of the curve, 8

$0/0$ indeterminate limiting form, not an arithmetic value, 11